HW5

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2026-03-03

Problem 1: Best subset selection

The data for this problem comes from a study by Stamey et al. (1989). They examined the relationship between the level of prostate-specific antigen and a number of clinical measures in men who were about to receive a radical prostatectomy. The variables are log cancer volume (lcavol), log prostate weight (lweight), age, log of the amount of benign prostatic hyperplasia (lbph), seminal vesicle invasion (svi), log of capsular penetration (lcp), Gleason score (gleason), and percent of Gleason scores 4 or 5 (ppp45). The last column corresponds to which observations were used in the training set and which were used in the test set (train).

Read in the prostate data set using the following code:

prostate = read.table('...Your file pathway..../prostate.data',header=TRUE)

Our response of interest here is the log prostate-specific antigen (lpsa). We will use this data set to practice 3 common subset selection approaches.

(a)

Approach 1: Perform best subset selection on the entire data set with lpsa as the response. For each model size, you will obtain a 'best' model (size here is just the number of predictors in the model): M1 is the best model with 1 predictor (size 1), M2 is the best model with 2 predictors (size 2), and so on.

Create a table of the AIC, BIC, adjusted R2 and Mallow's Cp for each model size. Report the model with the smallest AIC, smallest BIC, largest adjusted R2 and smallest Mallow's Cp. Do they lead to different results? Using your own judgement, choose a final model.

(b)

Approach 2: The dataset has already been split into a training and test set. Construct your training and test set based on this split. You may use the following code for convenience:

train = subset(prostate,train==TRUE)[,1:9]

test = subset(prostate,train==FALSE)[,1:9]

For each model size, you will obtain a 'best' model. Fit each of those models on the training set. Then evaluate the model performance on the test set by computing their test MSE.

Choose a final model based on prediction accuracy. Fit that model to the full dataset and report your final model here.

(c)

Approach 3 (practice): This approach is used to select the optimal size, not which pre- dictors will end up in our model. Split the dataset into k folds (you decide what k should be). We will perform best subset selection within each of the k training sets. Here are more detailed instructions:

i. (practice)For each fold k = 1, . . . , K:

1. Perform best subset selection using all the data except for those in fold k (training set). For each model size, you will obtain a 'best' model.
2. For each 'best' model, evaluate the test MSE on the data in fold k (test set).
3. Store the test MSE for each model.

Once you have completed this for all k folds, take the average of your test MSEs for each model size. In other words, for all k models of size 1, you will compute their k- fold cross-validated error. For all the k models of size 2, you will compute their k-fold cross-validated errors, and so on. Report your 8 CV errors here.

ii. (practice)

Choose the model size that gives you the smallest CV error. Now perform best subset selection on the full data set again in order to obtain this final model. Report that model here. (For example, suppose cross-validation selected a 5-predictor model. I would perform best subset selection on the full data set again in order to obtain the 5-predictor model.)

Problem 2: Concept Review

(a)

For the following statement, state whether or not it is True or False. Briefly justify your answer.

Suppose I have 3 models to pick from:

MA :Y ~ X1 + X2 + X3 + X4 + X5

MB :Y ~ X6 + X7 + X8 + X9 + X10

MC :Y ~ X1 + X2 + X7 + X9 + X10

Using AIC, BIC, Mallow's Cp, adjusted R2 could lead us to pick different final models.

(b)

For the following statement, state whether or not it is True or False. Briefly justify your answer.

Suppose I have two models:

M3 :Y ~ X1 + X2 + X4

M4 :Y ~ X4 + X5 + X6 + X7

It must be that RSS3 >= RSS4.

(c)

For the following statement, state whether or not it is True or False. Briefly justify your answer.

Suppose I have two models:

M2 :Y ~ X4 + X5

M4 :Y ~ X4 + X5 + X6 + X7

It must be that RSS2 >= RSS4.

(d)

(practice) What advantages are there to using AIC/BIC instead of using the test MSE as

Problem 3: Diagnostics for MLR (practice)

(a) List the four assumptions we make when fitting a multiple linear regression model.

(b) True or False? In order to fit a multiple linear regression model (i.e. obtain least square estimates), we must have distributional assumptions on the random error term epsilon i. Briefly justify your answer.

(c) True or False? The summary(lm()) output automatically gives you p-value for individual hypothesis tests: H0 : beta_j = 0 versus H1 : beta_j = 0, j = 1, . . . , p. These hypothesis tests assume that the random error term are normally distributed. Briefly justify your answer.

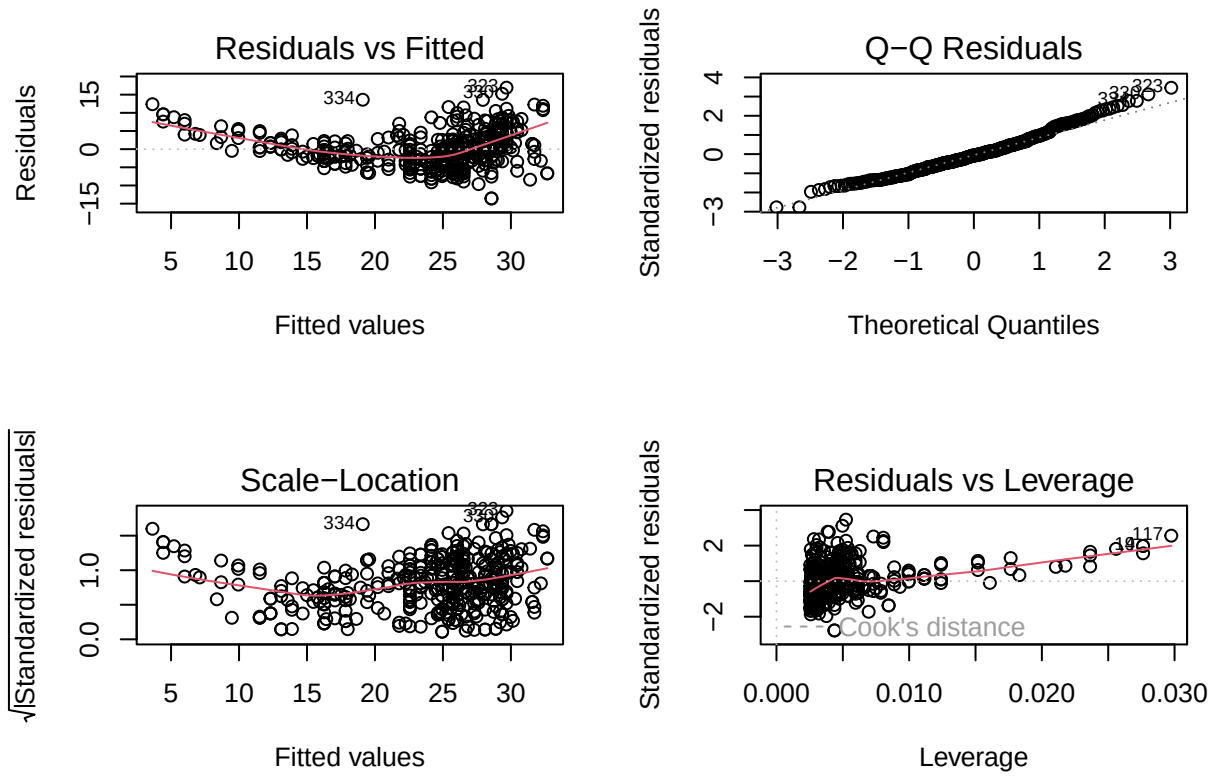
(d) Suppose we want to run some visual diagnostics to check whether or not the linearity as- sumption holds. Besides the fact that it is is time consuming, explain why it is not sufficient to check every pairwise scatterplot between Y and Xj , j = 1, . . . , p.

(e) To check whether or not the linearity assumption holds, we can run some diagnostics. Let's use the Auto dataset as an example from library(ISLR2). Run the following code:

```r
library(ISLR2)
m1 = lm(mpg~horsepower,data=Auto)
summary(m1)
```

```
##
## Call:
## lm(formula = mpg ~ horsepower, data = Auto)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -13.5710  -3.2592  -0.3435   2.7630  16.9240
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 39.935861   0.717499   55.66   <2e-16 ***
## horsepower  -0.157845   0.006446  -24.49   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.906 on 390 degrees of freedom
## Multiple R-squared:  0.6059, Adjusted R-squared:  0.6049
## F-statistic: 599.7 on 1 and 390 DF,  p-value: < 2.2e-16
```

```r
par(mfrow=c(2,2))
plot(m1)
```



Take a look at the plot in the top left corner (titled Residuals vs Fitted). This is what we call a residual plot because it is literally a plot of residuals versus fitted values (ˆyi). The red line is a smooth fit to the residuals, which is displayed in order to make it easier to identify any trends. You should observe that the plot exhibits a clear U-shape, which provides a strong indication of non-linearity in the data.

Ideally what we want to see is no discernible pattern in the residual plot and a relatively flat red line. A residual plot exhibiting no pattern and a flat red line indicates that the linearity assumption for our model holds. Propose a more flexible model than m1 that might be able to accommodate some non-linearity. Present your proposed model (call it m2) and its corresponding residual plot here. Is it an improvement over m1?

We'll discuss the remaining plots from plot(m1) and their utility in an upcoming lecture.